GAS IN EU INDUSTRY

# Gas, Wind & the Industrial Energy Fight

*Volumes, costs, the 2022 price shock, and why industry is harder to decarbonise than power generation.*

| ~40% | €340 | 60% | €220 |
|---|---|---|---|
| of EU industrial energy is gas | per MWh — crisis peak Aug 2022 | of industrial energy is hard to abate | green premium for ammonia today |

MA500 Capstone: Sustainability Consulting Project · Spring 2026

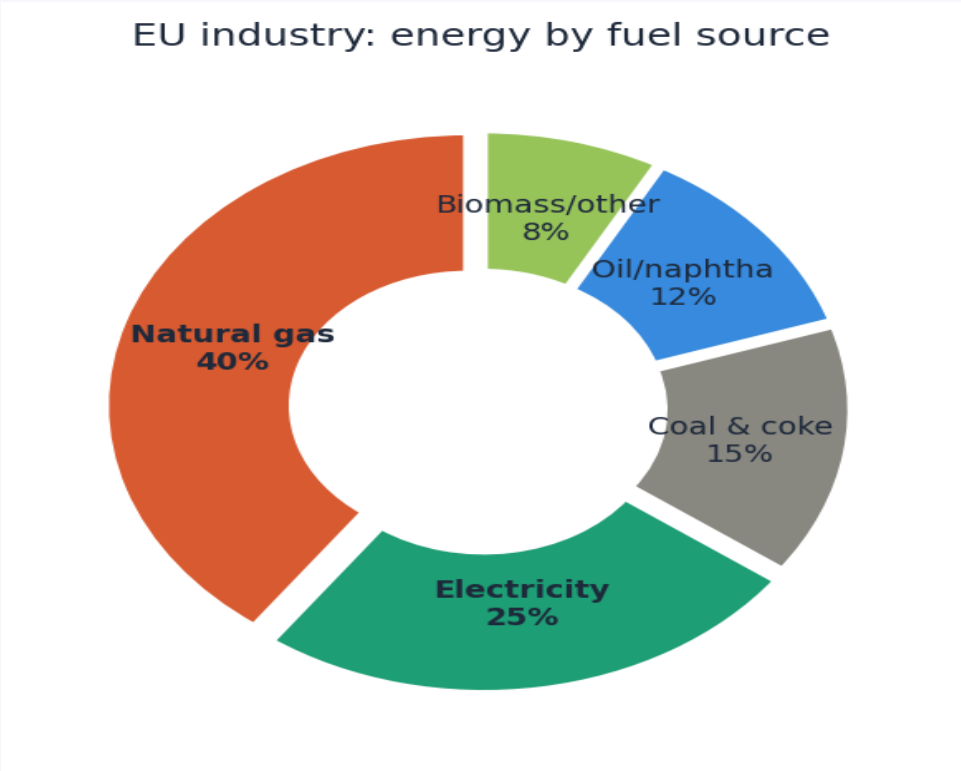
## THE VOLUME PICTURE — WHAT DOES GAS ACTUALLY DO IN EU INDUSTRY?

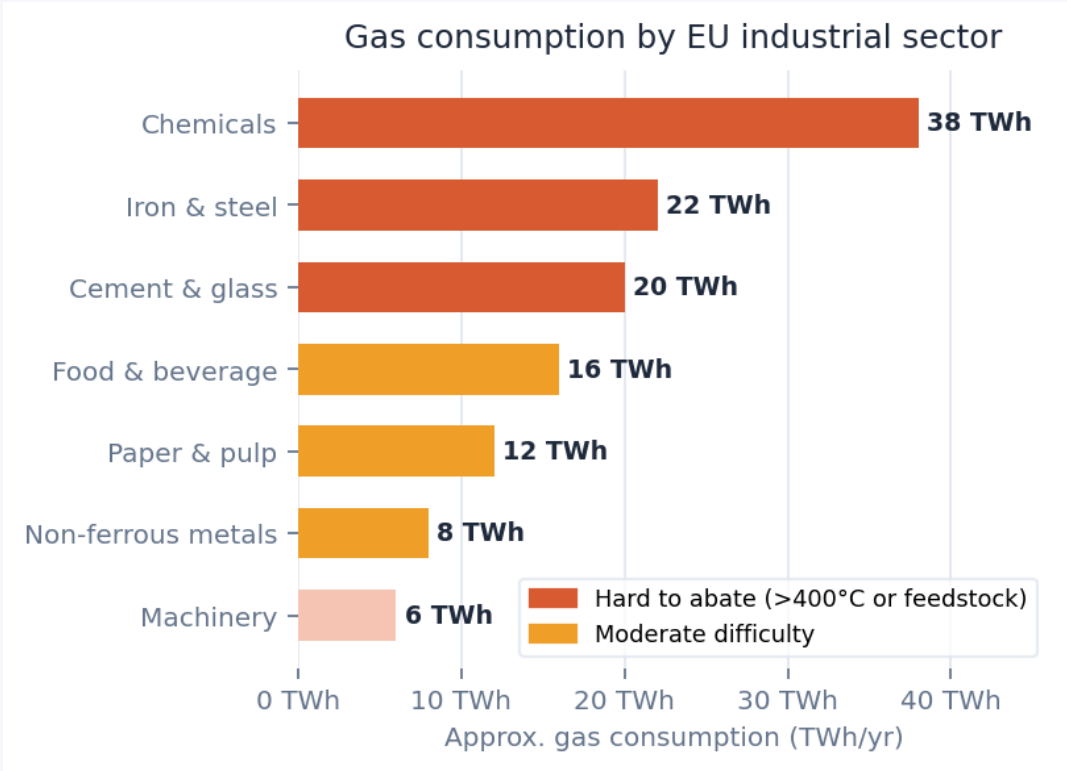
**~400 TWh**
EU industry total final energy / yr

**40%**
Gas share — single largest fuel source

**25%**
Electricity share — mostly motors

**60%**
Share requiring $H_2$ or CCS, not just electrons




*Sources: Eurostat Energy Balances 2022; IEA Tracking Clean Energy Progress 2023.*

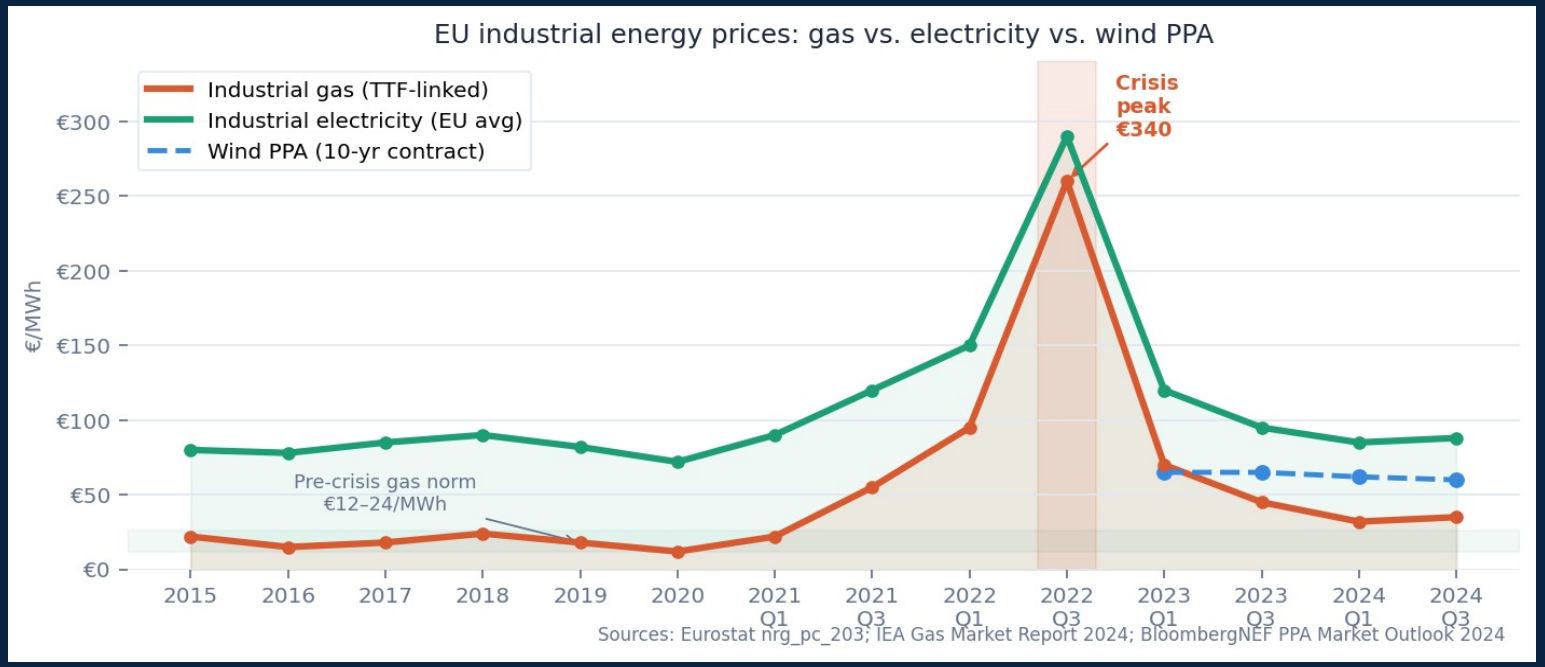
# THE PRICE STORY — GAS VS. ELECTRICITY VS. WIND PPA, 2015–2024

**€18/MWh**
Gas — 2019 pre-crisis norm

**€340/MWh**
Gas — Aug 2022 crisis peak

**€35/MWh**
Gas — 2024 normalised

**€55–75**
Wind PPA (10-yr contract, 2024)




### The crisis lesson

When gas spiked to €340/MWh, industrial energy costs became an existential threat. German chemical plants curtailed output; fertiliser

### Why electrification looked attractive

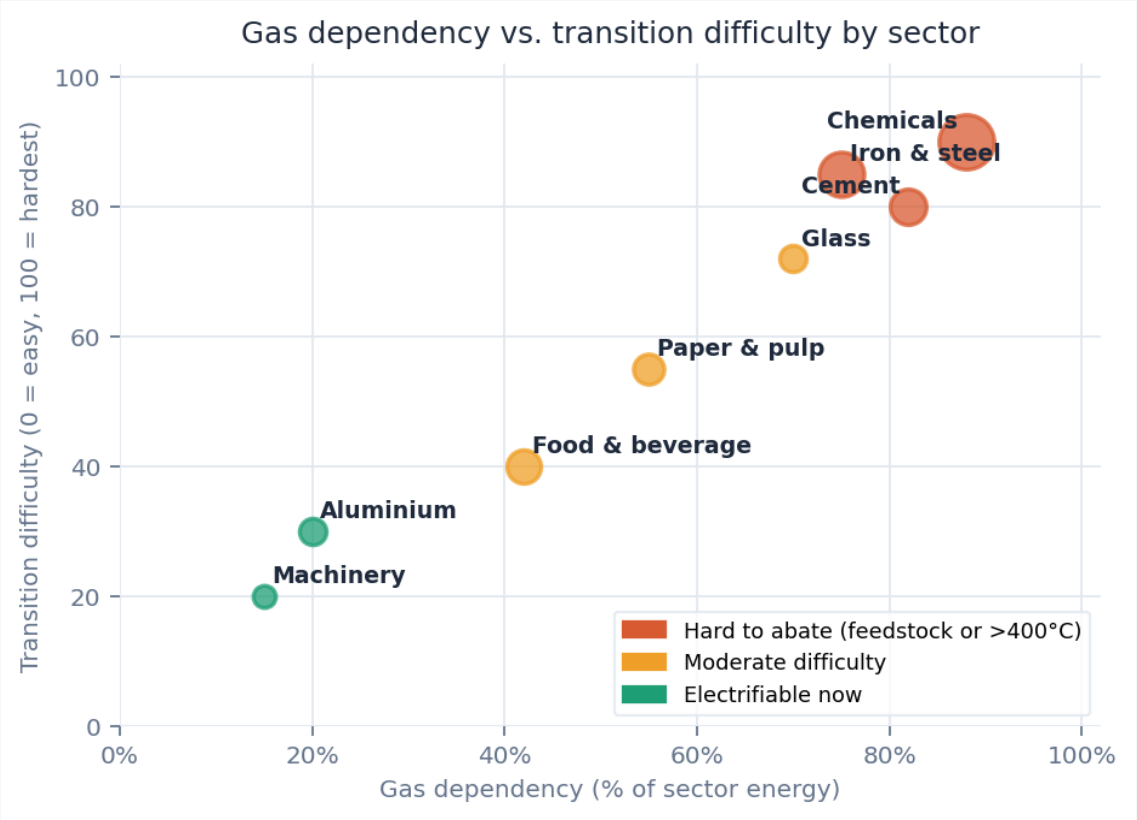
Fixed-price wind PPAs at €55–75/MWh looked extraordinarily cheap versus crisis gas — driving a wave of industrial PPA contracts in

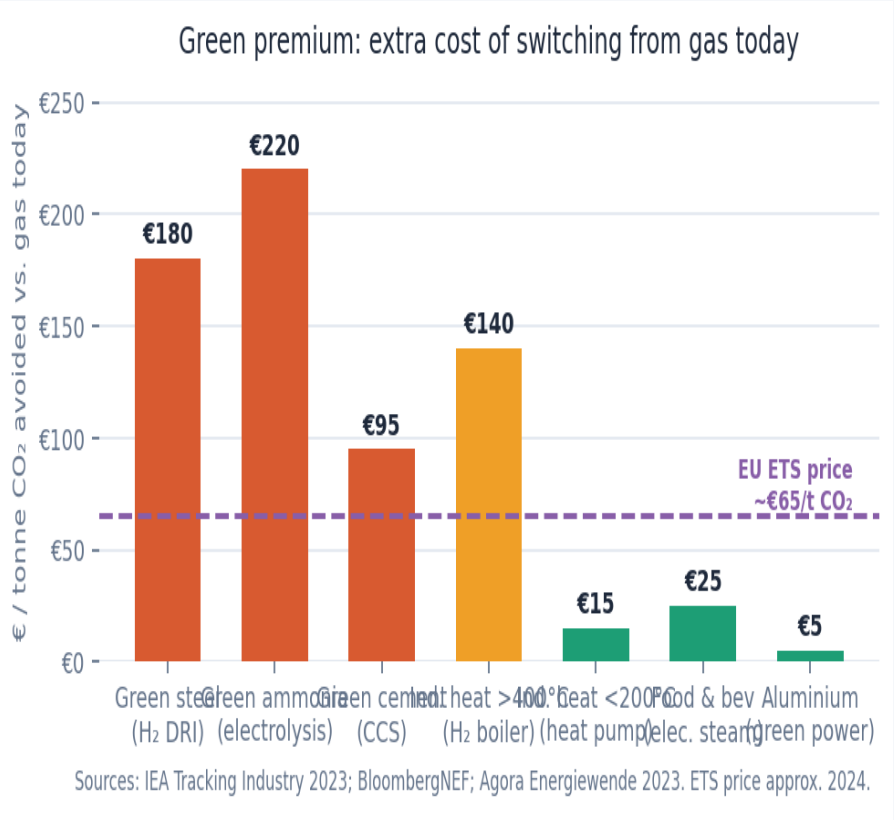
### The normalisation problem

With gas back at €35/MWh in 2024, industrial electricity at ~€85/MWh is again more expensive for direct heat. The crisis made the case; normalisation softened it.

## THE HARD-TO-ABATE PROBLEM — WHERE GAS CANNOT SIMPLY BE REPLACED

*The difficulty is not technological — it is thermodynamic and economic. High-temperature heat and chemical feedstocks cannot be replaced by electrons alone.*




*Sources: IEA Tracking Industry 2023; BloombergNEF Green Steel Tracker; Agora Energiewende Industrial Heat Report 2023. EU ETS price ~€65/t $CO_2$ (2024 avg).*

## STRATEGIC IMPLICATIONS — WHAT THIS MEANS FOR ØRSTED AND TOTALENERGIES

*Wind wins in power generation. The real contest — and the real prize — is industrial energy. That fight will be decided by green hydrogen costs, carbon pricing, and who controls industrial offtake contracts.*

### ØRSTED — The industrial opportunity

**Industrial PPAs are the new frontier**
As offshore wind scales, long-term industrial power purchase agreements with energy-intensive manufacturers are becoming a core revenue pillar — and a hedge against merchant price risk.

**Green hydrogen is the long game**
Offshore wind + electrolysis = green hydrogen. Ørsted is investing in Power-to-X. If green $H_2$ costs fall on a Wright's Law curve, Ørsted could own the supply chain for hard-to-abate sectors.

**The timing mismatch risk**
Chemicals and steel companies need supply certainty before they retrofit plants. Ørsted's project delays and cost overruns create credibility risk precisely when industrial customers need commitment.

**The scale required is enormous**
Replacing EU industrial gas with green hydrogen would require ~800–1,000 GW of offshore wind — roughly 10× current EU installed capacity. The market is vast; the timeline is long.

### TOTALENERGIES — The last stronghold and its limits

**Industrial gas demand is their best remaining argument**
Power generation gas is already losing. But hard-to-abate sectors need gas — as feedstock, as high-temperature heat — for longer. TotalEnergies' LNG book is defensible here, for now.

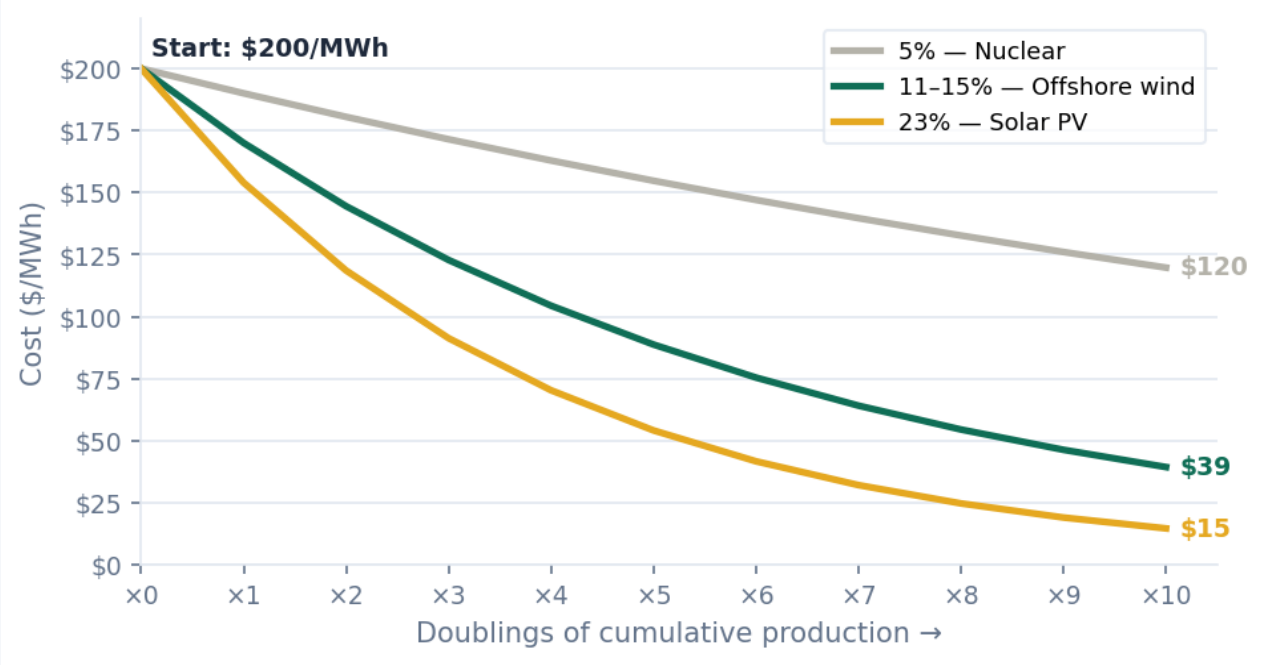
**Carbon pricing closes the window**
The EU ETS at €65/t $CO_2$ already makes green steel economics viable at scale. At €100–130/t — the IEA's NZE trajectory for 2030 — green ammonia crosses over too. The window is 5–10 years.

**Blue hydrogen as a bridge play**
TotalEnergies is positioning in CCS and blue hydrogen (gas + carbon capture) as a transition route for industrial decarbonisation. This buys market position without abandoning gas infrastructure.

**The capital allocation test**
Every euro of capex that goes into new LNG infrastructure is a euro not building the electrolyser and hydrogen network that industrial customers will eventually need. The choice is being made now.

*Consulting hypothesis: the industrial energy transition will be decided by green $H_2$ economics and EU ETS price trajectory — not by the offshore wind learning curve alone.*

# WRIGHT'S LAW — WHY COST KEEPS FALLING




## The rule

Every time cumulative production doubles, cost falls by a fixed percentage — the learning rate. It doesn't matter whether that doubling takes 2 years or 10.

## Why it compounds

Each cost reduction makes the technology more competitive → more deployment → another doubling → another cost cut. It is self-reinforcing.

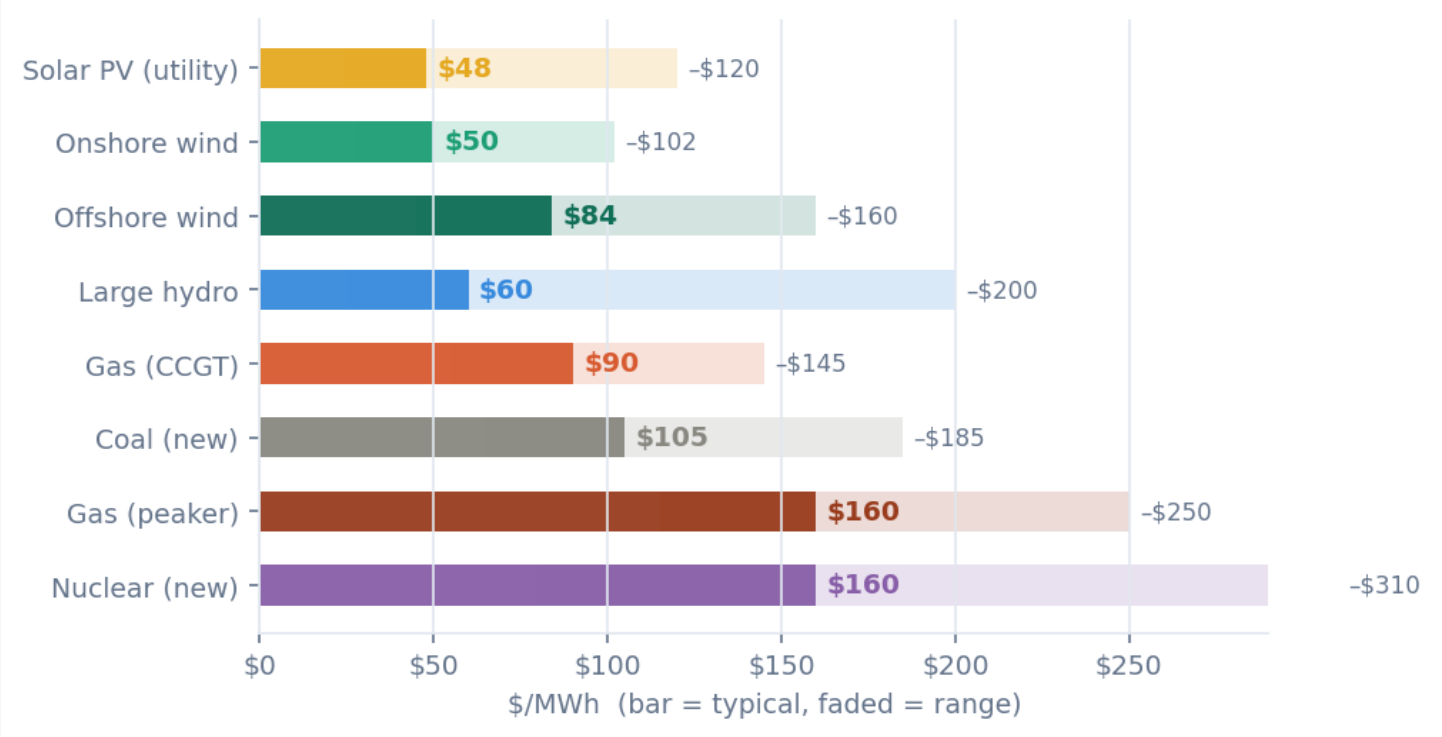
## Deployment IS the R&D

Technologies that get deployed at scale get dramatically cheaper. Those that don't — like nuclear — stagnate. Policy that drives deployment drives cost reduction.

*Offshore wind learning rate: ~11–15% per doubling. Solar PV: ~23%. Nuclear: effectively 0% or negative.*

# LCOE BY TECHNOLOGY — GLOBAL WEIGHTED AVERAGE, 2024

*Levelized Cost of Energy: the lifetime average cost per MWh — construction, financing, fuel and operations combined.*




**Onshore wind & solar**

Already cheaper than new-build gas in most markets — without any subsidy.

**Offshore wind**

Still above gas on mid LCOE — but the gap is closing, and gas carries fuel price volatility that LCOE doesn't capture.

**Nuclear (new)**

$80–$310/MWh. Widest uncertainty of any technology, and the only major source with a flat or negative learning curve.

*Sources: IRENA Renewable Power Generation Costs 2023; IEA WEO 2024. Bar = typical mid-point; faded range = best to worst market conditions.*

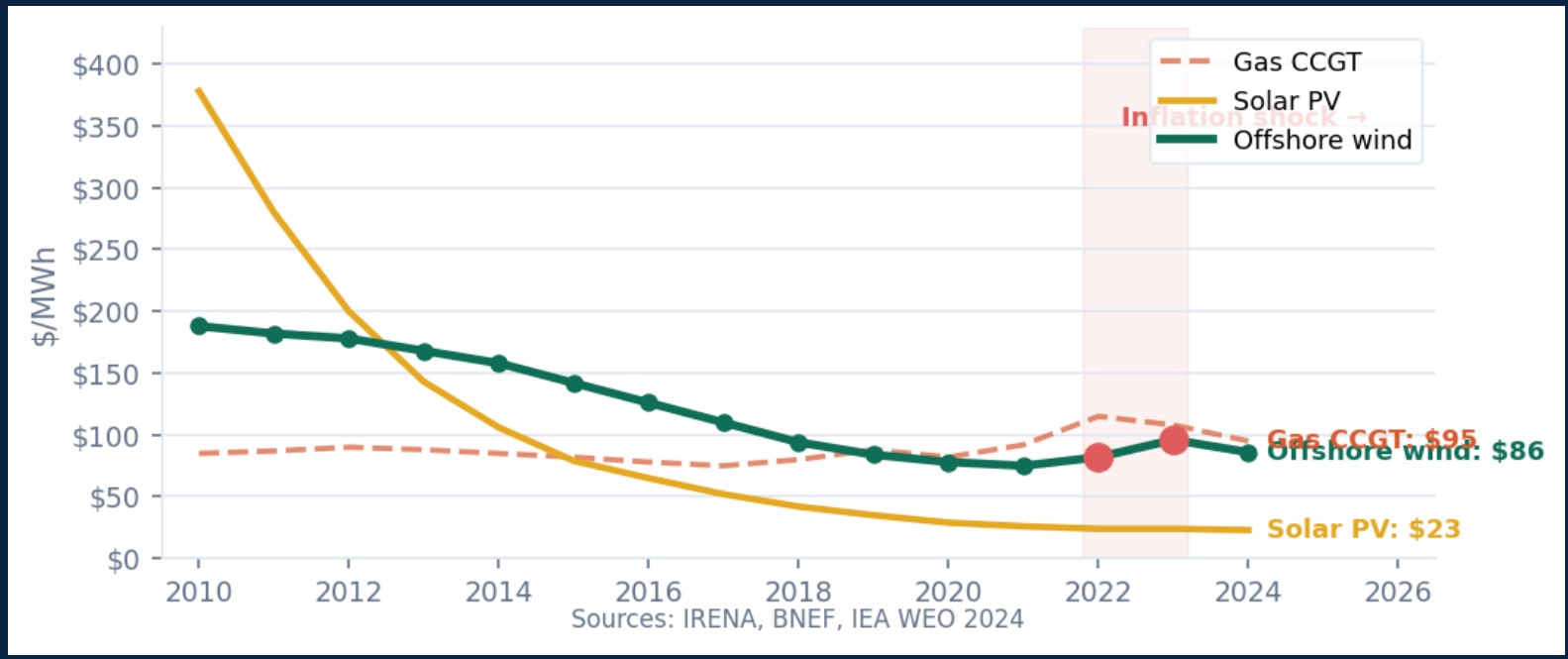
# OFFSHORE WIND COST TRAJECTORY — 2010 TO 2024

**$188**
2010
($/MWh)

**$75**
2021
($/MWh)

**$96**
Shock peak
2023

**~70%**
Total cost
reduction




### The long-run curve

~70% cost reduction over 14 years driven by turbine scale, supply chain learning, and developer experience.

### 2022–23 shock

Inflation + rate rises + supply chain disruption. Ørsted wrote down ~$4bn on US projects locked at pre-inflation auction prices.

### 2024 recovery

The learning curve resumed. The interruption was cyclical — the structural trajectory was not broken.

# THE STRATEGIC TENSION: WHAT FALLING LCOE MEANS FOR ØRSTED & TOTALENERGIES

*As offshore wind LCOE falls, the economic case for new fossil investment weakens — but the transition risk and stranded asset exposure differ fundamentally between these two companies.*

## ØRSTED — Riding the curve

### Wright's Law IS the business model

Ørsted bet its entire identity on the learning curve continuing. Every cost reduction validates the pivot from fossil fuels and enables new projects to clear their hurdle rate.

### A reinforcing feedback loop

Lower LCOE → more projects viable → more scale → lower LCOE. Ørsted's pipeline growth is both cause and consequence of cost reduction.

### 2022–23 exposed the execution risk

When inflation hit, projects locked at low fixed auction prices became loss-making overnight. The $4bn US write-down showed LCOE trajectory ≠ project profitability.

### The open question

Can Ørsted sustain its pipeline as it moves into deeper waters and less favorable markets — where the learning curve has less history to draw on?

## TOTALENERGIES — Straddling the divide

### Falling LCOE erodes the fossil case

As renewables get cheaper, the relative economics of LNG worsen — even without new carbon policy. Industrial buyers increasingly prefer fixed-price renewable PPAs over volatile gas contracts.

### Stranded asset risk is real

LNG infrastructure is a 20–30 year investment. If the LCOE curve continues, assets being sanctioned today may be uncompetitive before they are fully paid off.

### The hedging argument

TotalEnergies argues gas is a 'transition fuel' needed while renewables scale. This holds today; whether it holds to 2035 depends on how fast the LCOE curve continues.

### The core strategic question

Can TotalEnergies grow its renewables arm proportionally to falling LCOE — or will fossil returns continue to crowd out renewable capex in internal capital allocation?

*Consulting hypothesis: Ørsted's risk is execution at scale in adverse markets. TotalEnergies' risk is a strategic identity that cannot survive if the LCOE curve continues for another decade.*